



Exponential Equations and Graphs

Exponential equations are a powerful way to model growth and decay.

Solving Exponential Equations

Definition

An **exponential equation** contains a variable in an exponent.

To solve an exponential equation, isolate the term with the variable exponent. Use properties of exponents and transformations to make the terms on each side of the equals sign have equal bases. Then compare the exponents.

EXAMPLE 1

Solve.

A $2\sqrt{2} = 4^x$

SOLUTION

Transform each side of the equation so that 2 is the base. Then use the laws of exponents to simplify.

$$\begin{aligned} 2\sqrt{2} &= 4^x \\ 2^1 \cdot 2^{\frac{1}{2}} &= (2^2)^x && \text{Rewrite the left side using exponents.} \\ 2^{\frac{3}{2}} &= 2^{2x} && \text{Apply properties of exponents.} \\ \frac{3}{2} &= 2x && \text{Set the exponents equal to each other.} \\ \frac{3}{4} &= x && \text{Divide each side by 2.} \end{aligned}$$

► **Remember** e is a mathematical constant, approximately equal to 2.71828. It is not a variable.

B $e^{3-x} = 1$

SOLUTION

Transform each side of the equation so that e is the base. Then use the laws of exponents to simplify.

$$e^{3-x} = 1$$

$$e^{3-x} = e^0 \quad \text{Rewrite 1 as a power of } e.$$

$$3 - x = 0 \quad \text{Set the exponents equal to each other.}$$

$$3 = x \quad \text{Addition Property of Equality} \quad \blacksquare$$

► **Remember** For any positive number p , $p^0 = 1$.

Graphing Exponential Functions

How to Graph an Exponential Function

To graph an equation of the form $f(x) = ab^x + k$,

Step 1 Determine the horizontal asymptote $y = k$.

Step 2 Plot the y -intercept $(0, a + k)$.

Step 3 Plot two other points on the graph.

Step 4 Draw a smooth curve connecting the points.

EXAMPLE 2

$$f(x) = 2 \cdot 3^x - 1$$

A Graph the function.

SOLUTION

Step 1 Determine the horizontal asymptote.

$$y = -1$$

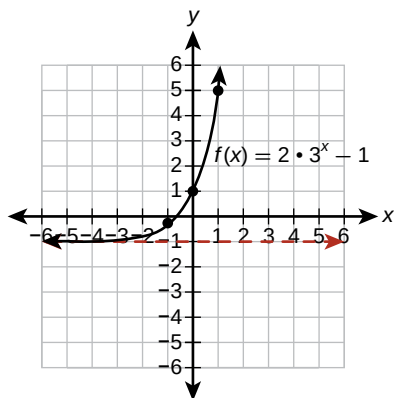
Step 2 Plot the y -intercept.

$$(0, 2 + (-1)) = (0, 1)$$

Step 3 Plot two other points on the graph.

$$\left(-1, -\frac{1}{3}\right) \text{ and } (1, 5)$$

Step 4 Draw a smooth curve connecting the points.



B Determine the domain and range.

SOLUTION

From the graph, you can see that x can take on any value; therefore, the domain is all real numbers. The graph also shows that the y -values are all greater than -1 , so the range is $y > -1$. ■

Determining an Exponential Function

You can write an exponential function to match a situation by determining the parameters a , b , and k .

EXAMPLE 3

What is an exponential function of the form $f(x) = ab^x + k$ that contains the points $(0, 4)$ and $(1, 5)$ and has a horizontal asymptote of $y = 2$?

► **Think About It** $f(0)$ gives the value for the y -intercept.

SOLUTION

Step 1 Use the horizontal asymptote to determine k .

The horizontal asymptote is $y = k$, and the problem states that $y = 2$.
So $k = 2$.

Step 2 Use the coordinates of the given points to determine a and b .

Substitute the values of k and the y -intercept of $(0, 4)$ into the function $f(x) = ab^x + k$. Solve for a .

$$4 = a \cdot b^0 + 2$$

$$4 = a + 2$$

$$a = 2$$

Substitute the values of a , k , and a different point, such as $(1, 5)$, into the function $f(x) = ab^x + k$. Solve for b .

$$5 = 2 \cdot b^1 + 2$$

$$3 = 2b$$

$$b = \frac{3}{2}$$

Step 3 Substitute the values of a , b , and k into $f(x) = ab^x + k$.

$$f(x) = 2 \cdot \left(\frac{3}{2}\right)^x + 2 \quad \blacksquare$$